

INSTRUCTOR GUIDE

Networking Capstone

22 phases · 4 parts · phase-by-phase coaching

This guide walks the capstone one phase at a time. For each phase you get the objective, the exact actions and commands the student is expected to produce, how to coach them through it, and the specific mistake to watch for.

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Running the Lab

Every student gets different numbers

Each student enters their last name plus computer number on the start screen. That string seeds their entire lab: VLAN IDs, IP subnets, gateway addresses, WAN circuit addressing, SSID, wireless passphrase, channel, and every credential. Two students sitting next to each other will have genuinely different correct answers.

► *The practical effect: copying a neighbour's screen produces a wrong answer that fails validation. Students must read their own Binder. Lean on this — it converts an anti-cheat mechanism into documentation practice.*

The Binder is the documentation

Tab opens the Binder at any time. It is a floating panel, so students can leave it open while they work in a terminal. Tabs cover the addressing plan, cable run schedule, circuit letter, wireless sheet, and credentials. When a student asks you for a value, point at the Binder rather than reading it out.

Phase gating and checkpoints

Phases validate continuously — a phase turns green the moment every task in it is satisfied, and the lab advances on its own. At the end of each of the four parts the lab stops and requires instructor initials before the student may continue.

- Any non-empty text is accepted as initials. It is an acknowledgement, not a password.
- The elapsed time for that part is recorded at the moment the checkpoint fires.
- Initials and per-part times both appear on the final completion report.

► *Use the checkpoint as a teaching pause, not a rubber stamp. The four gates land at Phase 8, Phase 15, Phase 19, and Phase 22 — each is a natural point to ask the student to explain what they just built.*

Completion report

After the final checkpoint is signed, the lab generates a completion report showing per-part times, the instructor initials for each part, totals, a timestamp, and a verification code derived from the student ID, times, and initials. Students can copy it for the LMS or download it as a text file. Because the code is derived from the contents, a retyped or edited report will not match a regenerated one for the same student ID.

Instructor console

Ctrl+Shift+I opens the instructor console. It asks for the passphrase every time it is opened. From there you can jump the lab to any phase, which builds the correct state for everything before it.

- Use it to recover a student who has lost work, or to demonstrate a later phase to the class.
- Parts skipped entirely are auto-signed with the initials INS so the report still renders.
- Jumping to the last phase of a part triggers that part's checkpoint normally.

► *The passphrase is set near the top of the HTML file and is meant to be changed before the file is distributed to students.*

Reading failures

When a ping fails, the lab prints a trace line that names the layer that broke — a missing cable, a shut port, a wrong VLAN, a missing gateway, an unassociated laptop. This is the most useful coaching tool in the lab. Train students to read the trace and work that specific layer instead of re-entering commands at random.

The terminal also validates input. Malformed addresses — a double dot, an octet over 255, a non-contiguous subnet mask — are rejected with an invalid input error rather than silently accepted. If a student sets a credential that does not match their Binder, the terminal accepts it the way real IOS would but prints a capstone note saying the check will not pass until it matches.

► *Students can never lock themselves out: the Binder password is always accepted at a login prompt, even if they configured a typo.*

Phase Map

Pacing estimates are starting points to calibrate against your own class — they are not measured figures. Expect Part 1 to run long with students who have never terminated cable and Part 3 to run long with everyone.

#	Title	Part	Gate
Phase 1	1st Floor Rack (IT Room)	Part 1	
Phase 2	2nd Floor Rack (Office 2 Closet)	Part 1	
Phase 3	Ground Floor End Devices	Part 1	
Phase 4	Second Floor End Devices	Part 1	
Phase 5	Inter-Closet Link	Part 1	
Phase 6	Ground Floor Cabling	Part 1	
Phase 7	Second Floor Cabling	Part 1	
Phase 8	Router & Carrier Handoff	Part 1	Checkpoint
Phase 9	AN-1 · Device Hardening	Part 2	
Phase 10	AN-1 · VLAN Database	Part 2	
Phase 11	AN-1 · Switchports	Part 2	
Phase 12	AN-2 · Device Hardening	Part 2	
Phase 13	AN-2 · VLAN Database	Part 2	
Phase 14	AN-2 · Switchports	Part 2	
Phase 15	Layer 2 Verification	Part 2	Checkpoint
Phase 16	CN-1 · Hardening	Part 3	
Phase 17	CN-1 · Router-on-a-Stick	Part 3	
Phase 18	CN-1 · WAN & Default Route	Part 3	
Phase 19	End-to-End Verification	Part 3	Checkpoint
Phase 20	Mount & Cable the WAP	Part 4	
Phase 21	Secure the WAP (IT1 → Browser)	Part 4	
Phase 22	Verify Wireless & Internet	Part 4	Checkpoint

Part 1 — Physical Layer

Racking, device placement, and structured cabling. Everything above Layer 1 depends on this being right, and nothing in this part tells the student when it is wrong. Mistakes here surface in Phase 19.

Phase 1 — 1st Floor Rack (IT Room)

Objective: Populate the ground-floor rack in the correct physical order.

WHAT THE STUDENT DOES

1. Walk to the IT Room rack and press E to open the rack panel.
2. Open the Equipment Box, take each device, and mount it into the numbered slot.
3. Rename each device as you rack it — the name is graded, not just the type.
4. Slot 1: PDU named PDU1. Slot 2: Patch Panel named Patch1. Slot 3: Switch named AN-1. Slot 4: Core Router named CN-1. Slot 5: Server named IT2.

HOW TO COACH IT

This phase is about rack discipline, not networking. Ask why the PDU goes at the top and the server at the bottom before you give the answer: power distribution feeds down, and the heaviest gear rides lowest so the rack does not become top-heavy. Students who can explain it will not forget it.

Naming is the first place students get sloppy. The validator compares the name exactly, so a device racked correctly but named "AN1" or "an-1" stays red. When a student says "it is in the right slot and it is still not green," have them re-read the task line and check the name before you touch anything.

WATCH FOR

Right slot, wrong name. Also students placing gear in slot order they invented rather than the order in the task list.

Phase 2 — 2nd Floor Rack (Office 2 Closet)

Objective: Repeat the rack build in the upstairs closet.

WHAT THE STUDENT DOES

5. Take the stairs to the second floor and find the Office 2 closet rack.
6. Slot 1: PDU named PDU2. Slot 2: Patch Panel named Patch2. Slot 3: Switch named AN-2.

HOW TO COACH IT

This is deliberately a repeat with different names. Let them do it unassisted — it is a confidence phase and a check that Phase 1 actually stuck.

If a student flies through this, that is your signal they are ready to be pushed harder on the cabling phases. If they hesitate, the problem is usually navigation, not knowledge: show them the stairs once and move on.

WATCH FOR

Naming the upstairs gear with the downstairs names (Patch1 instead of Patch2).

Phase 3 — Ground Floor End Devices

Objective: Place the ground-floor user devices on their desks.

WHAT THE STUDENT DOES

7. Place a PC on the IT Desk named IT1.
8. Place a PC on the Reception Desk named Reception1.
9. Place a Printer on the Reception Desk named Reception2.

HOW TO COACH IT

Point out that IT1 is the machine they will use later for the WAP browser and for host-sourced pings. It is not scenery — it is a tool they will come back to in Phases 19, 21, and 22.

Reception2 being a printer matters: printers land in a user VLAN like any host, and students often assume "printer" means something special. It does not, and saying so now saves confusion in Phase 11.

WATCH FOR

Dropping the printer on the wrong desk, or placing a PC where a printer is called for.

Phase 4 — Second Floor End Devices

Objective: Place the upstairs user devices.

WHAT THE STUDENT DOES

10. Place a PC on the Office 1 Desk named Data1.
11. Place a PC on the Office 2 Desk named Data2.
12. Place a Printer on the Office 2 Desk named Data3.

HOW TO COACH IT

Ask which VLAN these three will end up in before they get there. If they can already answer "Data," they are building a mental model instead of following a checklist.

Data1 is the far-end target for the Phase 19 cross-switch ping. Flag it now so the verification step later feels like a payoff rather than a new task.

WATCH FOR

Office 1 versus Office 2 desk mix-ups.

Phase 5 — Inter-Closet Link

Objective: Connect the two switches with the correct cable type.

WHAT THE STUDENT DOES

13. Open AN-1, click Port 1, then walk to AN-2 and click Port 1.
14. Set the cable type to crossover before pulling the run.

HOW TO COACH IT

This is the first real teaching moment in the lab. Make them justify the cable type out loud: two switches are like devices, so transmit and receive land on the same pins and the pairs must be crossed. Do not accept "because the task said crossover."

Mention that modern gear auto-negotiates with Auto-MDIX and this would work with a straight-through in the real world — then explain that the capstone grades the correct cable because they will be tested on the standard, and because they will meet gear old enough to care.

WATCH FOR

Pulling a straight-through and not understanding why it stays red. The cable colour is the tell: blue is straight, orange is crossover.

Phase 6 — Ground Floor Cabling

Objective: Wire the ground floor, including the structured-cabling path through the patch panel.

WHAT THE STUDENT DOES

15. Direct runs: IT1 → AN-1 Port 2, and IT2 (server) → AN-1 Port 5.
16. Desk drops: Reception1 → Reception Mount Port 1, Reception2 → Reception Mount Port 2.
17. Punch-downs (rear, permanent): Reception Mount Punch 1 → Patch1 Punch 1, and Punch 2 → Patch1 Punch 2.
18. Patch cords (front, replaceable): Patch1 Port 1 → AN-1 Port 3, and Patch1 Port 2 → AN-1 Port 4.

HOW TO COACH IT

This is the phase that teaches structured cabling, and it is where the most students stall. Draw the path on the whiteboard before they start: wall jack → punch-down block → patch panel front → switch. Four segments, one signal path.

Make the punch versus patch distinction explicit. The punch-down side is permanent copper run through walls and ceilings — you do not repull it because a port died. The front patch cord is the field-replaceable part. That is why the capstone colours punch runs violet and patch cords blue: they are different classes of cable with different lifecycles.

The internal bridge is one-to-one: Port 1 maps to Punch 1, Port 2 to Punch 2. A student who punches Mount Punch 1 to Patch1 Punch 2 has built a path that looks plausible and does not work. Have them trace it by number, not by eye.

WATCH FOR

Crossed punch numbering (Punch 1 to Punch 2). Also students trying to cable the wall mount jack straight to the switch and skipping the panel entirely.

Phase 7 — Second Floor Cabling

Objective: Repeat the structured path upstairs with three drops.

WHAT THE STUDENT DOES

19. Desk drops: Data1 → Office 1 Mount Port 1, Data2 → Office 2 Mount Port 1, Data3 → Office 2 Mount Port 2.
20. Punch-downs: Office 1 Mount Punch 1 → Patch2 Punch 1, Office 2 Mount Punch 1 → Patch2 Punch 2, Office 2 Mount Punch 2 → Patch2 Punch 3.
21. Patch cords: Patch2 Port 1 → AN-2 Port 2, Port 2 → AN-2 Port 3, Port 3 → AN-2 Port 4.

HOW TO COACH IT

Note the asymmetry: two different wall mounts feed one patch panel, and the punch numbering on the panel runs 1-2-3 while the mount numbering restarts per mount. This is exactly how a real closet looks and it is the point of the phase.

This is a good place to step back and let them fail for a few minutes. If a run is wrong they will find out in Phase 19 when a ping dies, and the trace output will name the layer. Students who debug their own Phase 7 mistake in Phase 19 learn more than students you rescued here.

WATCH FOR

Off-by-one punch numbering on Patch2, since three drops feed it from two mounts.

Phase 8 — Router & Carrier Handoff

Objective: Uplink the router and land the carrier circuit.

WHAT THE STUDENT DOES

22. Trunk uplink: CN-1 Fa0/0 → AN-1 Port 6.
23. Carrier handoff: CN-1 Fa0/1 → DEMARC Port 1.
24. Read the circuit letter posted on the NID before wiring the handoff.

HOW TO COACH IT

Fa0/0 is the router-on-a-stick uplink: one physical cable that will carry every VLAN once it is trunked in Phase 11 and subinterfaced in Phase 17. Say that out loud now so Phase 17 is not a surprise.

The DEMARC is the demarcation point — the boundary of responsibility. Everything on the provider side is not theirs to configure, ever. Airmen who understand demarcation stop trying to fix problems that belong to the circuit owner.

Send them to read the circuit letter now even though they do not need the addressing until Phase 18. Building the habit of reading the handoff documentation before touching the WAN is the lesson.

WATCH FOR

Cabling Fa0/1 to the switch and Fa0/0 to the DEMARC — reversed. Nothing complains until Phase 18.

CHECKPOINT — end of Part 1. The lab stops here and requires your initials before the student continues.

Part 2 — Layer 2

Device hardening, VLAN databases, access ports, and trunks on both switches. This is where the CLI work begins and where the seeded values start mattering.

Phase 9 — AN-1 · Device Hardening

Objective: Bring AN-1 to the baseline security standard.

WHAT THE STUDENT DOES

25. Open the AN-1 rack panel and click Console to open the terminal.

26. Set hostname, enable secret, console and VTY passwords with login, password encryption, the DoD banner, and disable DNS lookup.

COMMANDS

```
enable
configure terminal
hostname AN-1
enable secret <enable secret from Binder>
service password-encryption
no ip domain-lookup
banner motd #WARNING: DoD System#
line console 0
  password <console password>
  login
  exit
line vty 0 4
  password <vty password>
  login
  exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Every credential is seeded per student and lives in the Binder (Tab: Credentials). They cannot copy their neighbour. Insist they read their own Binder — that is the whole anti-cheat design and it doubles as documentation practice.

The banner is not decoration. Explain that without a proper warning banner, prosecuting unauthorised access becomes materially harder. This is one of the few places in the lab where the "why" is legal rather than technical, and it lands well with military students.

The lab has a safety valve: if they set a credential that does not match the Binder, the terminal accepts it like real IOS but prints a capstone note telling them it will not pass the check. Teach them to read that note instead of ignoring it. The Binder password is also always accepted at login prompts, so a typo can never lock a student out of their own device.

WATCH FOR

Setting a password on a line but forgetting login, which leaves the line unprotected. Also enable password instead of enable secret.

Phase 10 — AN-1 · VLAN Database

Objective: Build all five VLANs with the seeded IDs and names.

WHAT THE STUDENT DOES

27. From configure terminal, create each VLAN and name it exactly as the Binder specifies.
28. Create Dead_Vlan and set it to state suspend.

COMMANDS

```
vlan <MGMT ID>
  name <Management name>
  exit
vlan <RECEP ID>
  name <Reception name>
  exit
vlan <DATA ID>
```

```
name <Data name>
exit
vlan <NATIVE ID>
name <Native name>
exit
vlan <DEAD ID>
name Dead_Vlan
state suspend
exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

The VLAN IDs are seeded and will not match the student next to them. When two students compare screens and see different numbers, use it: ask which one is wrong. Neither is. That conversation makes the point that they are configuring to documentation, not to memory.

Dead_Vlan with state suspend is the parking lot for unused ports. Ask why you would suspend it rather than just leaving ports in VLAN 1. The answer — VLAN 1 is the default everything lands in, so an unconfigured port is a live port on a live VLAN — is a genuine security lesson.

Names are graded. "Management" and "MGMT" are not the same string.

WATCH FOR

Creating the VLAN but skipping the name, or forgetting state suspend on Dead_Vlan.

Phase 11 — AN-1 · Switchports

Objective: Assign access ports, build both trunks, and secure the unused ports.

WHAT THE STUDENT DOES

- 29. Access ports: Fa0/2 to Management, Fa0/3 and Fa0/4 to Reception, Fa0/5 to Management.
- 30. Trunks: Fa0/1 (to AN-2) and Fa0/6 (to CN-1), both with the allowed VLAN list and the native VLAN.
- 31. Unused: Fa0/7 and Fa0/8 to Dead_Vlan and shutdown.

COMMANDS

```
interface fa0/2
  switchport mode access
  switchport access vlan <MGMT ID>
  exit
interface range fa0/3 - 4
  switchport mode access
  switchport access vlan <RECEP ID>
  exit
interface fa0/5
  switchport mode access
  switchport access vlan <MGMT ID>
  exit
interface fa0/1
  switchport mode trunk
  switchport trunk allowed vlan <MGMT>,<RECEP>,<DATA>
  switchport trunk native vlan <NATIVE ID>
  exit
interface fa0/6
  switchport mode trunk
  switchport trunk allowed vlan <MGMT>,<RECEP>,<DATA>
```

```
switchport trunk native vlan <NATIVE ID>
exit
interface range fa0/7 - 8
switchport mode access
switchport access vlan <DEAD ID>
shutdown
exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

The allowed list is the three user VLANs only — Management, Reception, and Data. The native VLAN is carried untagged and is deliberately not in the allowed list. Students who add it anyway should be asked what "native" means before you correct them.

Two trunks matter for different reasons. Fa0/1 carries VLANs between the two switches. Fa0/6 carries every VLAN up to the router so router-on-a-stick can do inter-VLAN routing in Phase 17. If Fa0/6 is wrong, Part 3 fails and the cause is two phases behind them.

Teach interface range here. Doing Fa0/3 and Fa0/4 separately works, but range is the habit you want when they are configuring 24 ports on real gear.

Note for later: Fa0/7 gets shut down here and gets re-enabled in Phase 20 for the wireless AP. That is intentional. Unused ports get locked down, and when a port is genuinely put into service you re-open it on purpose.

WATCH FOR

Native VLAN included in the allowed list. Trunk configured on Fa0/6 but not Fa0/1, or vice versa. Access ports left in the default VLAN because switchport mode access was set but switchport access vlan was not.

Phase 12 — AN-2 · Device Hardening

Objective: Apply the same baseline to the upstairs switch.

WHAT THE STUDENT DOES

32. Console into AN-2 from the second-floor rack.
33. Same hardening block as Phase 9, with hostname AN-2.

COMMANDS

```
enable
configure terminal
hostname AN-2
enable secret <enable secret from Binder>
service password-encryption
no ip domain-lookup
banner motd #WARNING: DoD System#
line console 0
password <console password>
login
exit
line vty 0 4
password <vty password>
login
exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Same credentials as AN-1 — the Binder issues one set per student, applied to all three devices. Students sometimes expect a second set and go hunting.

This is a speed run for anyone who understood Phase 9. If a student is slow here, the issue is usually terminal navigation rather than concepts: check they are using exit and end correctly instead of closing and reopening the console.

WATCH FOR

Hostname left as AN-1 because they copied their own earlier commands wholesale.

Phase 13 — AN-2 · VLAN Database

Objective: Rebuild the full VLAN database on the second switch.

WHAT THE STUDENT DOES

34. Create all five VLANs again on AN-2 with the same seeded IDs and names.
35. Dead_Vlan again set to state suspend.

COMMANDS

```
vlan <MGMT ID>
  name <Management name>
  exit
vlan <RECEP ID>
  name <Reception name>
  exit
vlan <DATA ID>
  name <Data name>
  exit
vlan <NATIVE ID>
  name <Native name>
  exit
vlan <DEAD ID>
  name Dead_Vlan
  state suspend
  exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

This is the highest-value teaching moment in Part 2. Almost every student assumes the trunk they just built copies the VLAN database across. It does not. There is no VTP in this lab, and a trunk carries tagged frames — it does not replicate configuration.

Ask the question before they discover it: "You built a trunk to AN-2 last phase. Do you need to create the VLANs again?" Let them answer, then have them run show vlan brief on AN-2 to see the empty database for themselves. That beats being told.

The follow-on point: a VLAN that exists on one switch and not the other means traffic tagged for it gets dropped at the far end. That is precisely the failure they would chase for an hour on real gear.

WATCH FOR

Skipping this phase entirely on the assumption the trunk handled it.

Phase 14 — AN-2 · Switchports

Objective: Assign the upstairs access ports and match the trunk configuration.

WHAT THE STUDENT DOES

- 36. Access: Fa0/2, Fa0/3, and Fa0/4 to the Data VLAN.
- 37. Trunk: Fa0/1 to AN-1 with the same allowed list and native VLAN as the far side.
- 38. Unused: Fa0/5 through Fa0/8 to Dead_Vlan and shutdown.

COMMANDS

```
interface fa0/2
  switchport mode access
  switchport access vlan <DATA ID>
  exit
interface fa0/3
  switchport mode access
  switchport access vlan <DATA ID>
  exit
interface fa0/4
  switchport mode access
  switchport access vlan <DATA ID>
  exit
interface fa0/1
  switchport mode trunk
  switchport trunk allowed vlan <MGMT>,<RECEP>,<DATA>
  switchport trunk native vlan <NATIVE ID>
  exit
interface range fa0/5 - 8
  switchport mode access
  switchport access vlan <DEAD ID>
  shutdown
  exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Both ends of a trunk must agree on the native VLAN. Mismatched native VLANs are how traffic leaks between segments, and on real Cisco gear CDP will scream about it. Make them state the native VLAN on both switches out loud and confirm they match.

Ask why all three upstairs ports are Data while downstairs had a mix. The answer is that VLAN membership follows function and location, not port number — the upstairs offices are all Data users.

WATCH FOR

Native VLAN typo on one side only. This will not fail until Phase 19 and it fails confusingly.

Phase 15 — Layer 2 Verification

Objective: Read the switch state and save both configurations.

WHAT THE STUDENT DOES

- 39. On AN-1: show vlan brief, then show interfaces trunk.

40. On AN-2: the same two commands.
41. Save both: copy running-config startup-config.

COMMANDS

```
show vlan brief
show interfaces trunk
copy running-config startup-config
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Do not let them run the shows and move on. Make them read the output against the Binder line by line. The point of a verification phase is building the habit of confirming intent against reality — a student who runs show commands without reading them has learned nothing.

Good questions to ask at this gate: which ports are in Dead_Vlan and are they down? Does the trunk list show both trunks with the right native VLAN? Are all five VLANs present on both switches?

Saving matters. Unsaved configuration dies with the power, and this is the moment to tell the story of someone who spent a shift configuring a switch and lost it to a reboot.

WATCH FOR

Running the shows without reading them, and skipping the save on the second switch.

CHECKPOINT — end of Part 2. The lab stops here and requires your initials before the student continues.

Part 3 — Layer 3

Router hardening, router-on-a-stick inter-VLAN routing, the WAN circuit, and end-to-end verification from the endpoints.

Phase 16 — CN-1 · Hardening

Objective: Apply the baseline security standard to the router.

WHAT THE STUDENT DOES

42. Console into CN-1 from the ground-floor rack.
43. Same hardening block as the switches, with hostname CN-1.

COMMANDS

```
enable
configure terminal
hostname CN-1
enable secret <enable secret from Binder>
service password-encryption
no ip domain-lookup
banner motd #WARNING: DoD System#
line console 0
password <console password>
login
exit
line vty 0 4
password <vty password>
login
```

```
exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Same standard, different platform. Make the point that hardening is a policy applied to every managed device, not a switch-specific task. In the field this would be a template or a script.

This is a short phase. Use the spare minutes to preview router-on-a-stick, because Phase 17 is the conceptual peak of the lab.

WATCH FOR

Students assuming a router needs different hardening than a switch.

Phase 17 — CN-1 · Router-on-a-Stick

Objective: Create the VLAN gateways as subinterfaces on the single uplink.

WHAT THE STUDENT DOES

44. Bring up the physical interface Fa0/0 with no shutdown.
45. Create one subinterface per user VLAN, set dot1q encapsulation, and assign the gateway IP from the Binder.

COMMANDS

```
interface fa0/0
  no shutdown
  exit
interface fa0/0.<MGMT ID>
  encapsulation dot1q <MGMT ID>
  ip address <Management gateway> 255.255.255.0
  exit
interface fa0/0.<RECEP ID>
  encapsulation dot1q <RECEP ID>
  ip address <Reception gateway> 255.255.255.0
  exit
interface fa0/0.<DATA ID>
  encapsulation dot1q <DATA ID>
  ip address <Data gateway> 255.255.255.0
  exit
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Bringing up the physical interface first is not optional and students skip it constantly. Subinterfaces ride on the physical link — if Fa0/0 is administratively down, every gateway is down with it and nothing routes. Have them do no shutdown on Fa0/0 before anything else.

The concept to land: one physical cable, three logical gateways, separated by 802.1Q tags. The switch tags frames with the VLAN ID, the router reads the tag and hands the frame to the matching subinterface, and that subinterface holds the gateway address for that subnet. Draw it if you have a board.

The subinterface number and the encapsulation VLAN are two different things that students conflate. Convention is to number the subinterface to match the VLAN so a human can read it — but the encapsulation statement is what actually assigns the VLAN. Ask them which line would break routing if it were wrong. It is the encapsulation.

Gateway addresses come from the Binder addressing plan and are seeded per student. They are always the .1 of that VLAN subnet.

WATCH FOR

Forgetting no shutdown on Fa0/0. Encapsulation VLAN not matching the subinterface number. Using the wrong /24 for a gateway.

Phase 18 — CN-1 · WAN & Default Route

Objective: Address the carrier link and point unknown traffic at the provider.

WHAT THE STUDENT DOES

46. Read the circuit letter on the DEMARC for the customer-side and provider-side addresses.
47. Put the customer-side IP on Fa0/1 with a /30 mask and bring it up.
48. Add a default route toward the provider IP.

COMMANDS

```
interface fa0/1
ip address <customer side IP> 255.255.255.252
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 <provider side IP>
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

Explain the /30 before they type it: a point-to-point link needs exactly two usable addresses, so a 255.255.255.252 mask is the right-sized subnet. Ask how many usable hosts a /30 gives. If they say two, they understand it.

The default route is the "everything else" rule. Anything the router has no specific route for goes to the provider. This is the single line that turns a working LAN into a network with internet access, and it is worth pausing on.

Customer side versus provider side trips people up. The customer side goes on their interface; the provider side is the next hop in the route statement. Getting them backwards produces a router that cannot reach its own next hop.

The console now rejects malformed addresses. If a student fat-fingers a double dot or an octet over 255, they get an invalid input error instead of a silently broken config. Tell them this is the terminal helping them, and that real IOS does the same.

WATCH FOR

Customer and provider addresses swapped. Wrong mask (/24 instead of /30). Missing no shutdown on Fa0/1.

Phase 19 — End-to-End Verification

Objective: Prove the wired network works from the endpoints, not from the gear.

WHAT THE STUDENT DOES

49. From IT1 (walk to the desk, press E, open CMD): ping the Management gateway, then ping a Data host.
50. From Reception1: ping a Data host upstairs to prove the inter-switch trunk carries traffic.
51. From any PC: ping the provider side of the circuit to prove the default route and WAN.

52. On CN-1: show ip route to confirm three connected VLAN subnets and the default route.

53. Save CN-1 with copy running-config startup-config.

COMMANDS

```
ping <Management gateway>
ping <Data host IP>
ping <provider side IP>
show ip route
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

The pings must come from the PCs, not from the router. This is the difference between "my config looks right" and "the user can actually reach the resource." Real technicians verify from the endpoint, and students who skip this habit will close tickets that are not fixed.

When a ping fails, the lab prints a trace line naming the layer that broke — a missing cable, a shut port, a wrong VLAN, a missing gateway. Teach them to read it and work that layer instead of randomly re-entering commands. This is the single most valuable diagnostic habit in the whole capstone.

In show ip route, have them identify the three connected subnets (C) and the static default (S*). Ask what would happen if one connected route were missing — the answer points straight back at a subinterface or a shut port.

This gate is where Part 1 and Part 2 mistakes finally surface. If a student is failing here, resist fixing it. Ask which ping failed, what the trace said, and which phase built that piece. Walk them backwards through their own work.

WATCH FOR

Pinging from the router console because it is easier. Also students declaring success when only one of the three pings passes.

CHECKPOINT — end of Part 3. The lab stops here and requires your initials before the student continues.

Part 4 — Wireless

Mounting and cabling the access point through the patch panel, securing it through its web GUI, and proving guest wireless reaches the internet.

Phase 20 — Mount & Cable the WAP

Objective: Install the wireless access point and wire it through the patch panel.

WHAT THE STUDENT DOES

54. Open the Equipment Box, click the Wireless AP to pick it up, carry it to the Guest Lounge, and press E under the green ceiling-mount marker.
55. Rename the AP to match the Binder wireless sheet.
56. Cable the AP PoE/LAN jack down to an open Patch Panel data port.
57. Punch that panel port across the back to a spare port on the Reception/Guest switch.
58. On the switch, set that port to access mode in the Reception VLAN and no shutdown.

COMMANDS

```
interface fa0/7
switchport mode access
```

```
switchport access vlan <RECEP ID>
no shutdown
exit
copy running-config startup-config
```

► Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.

HOW TO COACH IT

The signal path is WAP → Patch Panel → Switch, and it is deliberately the same structured path they built for the desk drops in Phase 6. An access point is just another drop; the only difference is that it hangs from a ceiling instead of sitting on a desk. If they understood Phase 6, this is the same skill applied to new gear.

Fa0/7 was shut down into Dead_Vlan back in Phase 11. They must now deliberately re-open it. This is the whole lesson of the phase in one action: unused ports stay locked, and a port gets opened only when something real is plugged into it. Ask them why the port was closed before you let them open it.

The AP lands in the Reception VLAN, not a dedicated wireless VLAN, because guest wireless shares the reception segment in this design. If a student asks whether guest traffic should be separated, that is a good instinct — tell them yes in a production design, and that the Binder defines the design here.

Naming matters again: the AP name is graded against the Binder wireless sheet.

WATCH FOR

Cabling the AP straight to the switch and skipping the panel. Forgetting that Fa0/7 is still shut from Phase 11 and wondering why the wireless never comes up.

Phase 21 — Secure the WAP (IT1 → Browser)

Objective: Configure the access point through its web GUI from a workstation.

WHAT THE STUDENT DOES

59. Walk to IT1, press E, and open the Browser.
60. Navigate to the AP management IP from the Binder.
61. Change the admin password off its factory default.
62. Set the SSID, set security to WPA2-PSK with the Binder passphrase, disable WPS, and set the assigned channel.

HOW TO COACH IT

Note that this is a browser task, not a CLI task. Consumer and small-business APs are managed over HTTP, and students need to be comfortable moving between a terminal and a web GUI in the same job.

Five settings are graded and each maps to a real finding on a security scan: default admin credentials, SSID, WPA2 rather than open, WPS disabled, and the correct channel. Ask them which of the five they would flag first on an inspection. Default admin credentials is the right answer.

WPS deserves a moment. It exists for convenience and it is a known brute-forcible weakness — it is disabled on any network anyone cares about. This is a concrete example of a feature shipped on by default that a technician turns off.

The channel assignment is seeded to 1, 6, or 11. Ask why only those three. They are the non-overlapping 2.4 GHz channels in North America, and channel planning in a building with multiple APs is exactly this arithmetic.

WATCH FOR

Configuring the SSID and passphrase but leaving the admin password at default. Setting a channel other than the seeded one.

Phase 22 — Verify Wireless & Internet

Objective: Join the wireless network from the laptop and prove internet reachability.

WHAT THE STUDENT DOES

63. Take the Test Laptop from the Equipment Box and place it on the Lounge Table.
64. Open the laptop, join the SSID, and enter the passphrase from the Binder.
65. Open CMD on the laptop and ping the gateway and the provider side.
66. Open the Browser on the laptop and load RedWolf Search to prove internet access.

COMMANDS

```
ipconfig  
ping <Reception gateway>  
ping <provider side IP>
```

► *Angle brackets are seeded per student — the values come from that student's Binder, not from this guide.*

HOW TO COACH IT

This is the victory lap and the full-stack proof. A page loading on a wireless laptop means every layer they built is working: cabling, VLANs, trunks, routing, the default route, the AP configuration, and the wireless association. Say that explicitly — students rarely notice how much is riding on that one page load.

Have them run ipconfig before pinging. If the laptop shows an APIPA address or media disconnected, the association or the uplink is the problem, not routing. Reading the address before reaching for ping is the habit worth building.

If the browser will not load, walk the chain in order: is the laptop associated, is the AP powered and cabled, is the uplink port in the right VLAN and up, does that VLAN have a gateway, is the default route present. The failure text names the layer — make them use it.

When the page loads, the final checkpoint fires. Sign it and the completion report generates.

WATCH FOR

Joining the SSID with a mistyped passphrase. Also students trying the browser before checking association.

CHECKPOINT — end of Part 4. The lab stops here and requires your initials before the student continues.

Appendix A — Worked Example

These are the values for the seed RIVERA12. They are shown so you can see the shape of a completed Binder. Every student's numbers will differ — do not hand these out as answers.

VLANs

VLAN	ID	Name	Gateway
Management	10	Management	192.168.7.1
Reception	120	Reception	192.168.23.1
Data	30	Data	192.168.83.1
Native	380	Native	—
Dead	888	Dead_Vlan (suspend)	—

WAN circuit

Field	Value
Circuit ID	NIPR-7500-B
Provider side	10.101.242.1
Customer side	10.101.242.2
Mask	255.255.255.252 (/30)

Wireless

Field	Value
AP name	WAP1
SSID	GUEST-997
Passphrase	Lounge58!wpa
Channel	6
Management IP	192.168.1.50

Appendix B — Fast Diagnostics

When a student is stuck and you have thirty seconds, work down this list in order.

- Ping fails from a PC — read the trace line. It names the layer. Start there, not at the top.
- Nothing on one floor works — check the crossover between AN-1 Port 1 and AN-2 Port 1, then the trunk on both sides.
- One host only is unreachable — check its punch-down numbering. Port N bridges to Punch N, and a crossed pair looks correct at a glance.
- Inter-VLAN routing dead — check that Fa0/0 on CN-1 has no shutdown, then that each subinterface encapsulation matches its VLAN.
- Internet dead but LAN fine — check the default route and that Fa0/1 has the customer-side address, not the provider-side.
- A phase will not turn green and the config looks right — check the device name and the VLAN name against the Binder. Names are compared exactly.

- Wireless will not associate — check the passphrase, then that the AP uplink port is in the Reception VLAN and not still shut from Phase 11.
- Everything looks right and nothing works — have them run `show running-config` and read it aloud. They will find it themselves about half the time.